| Program : **Diploma in Textile Technology** | |
|---|---|
| Course Code : **3066** | Course Title: **Fabric manufacture I Lab** |
| Semester : **3** | Credits: **1.5** |
| Course Category: **Program core** | |
| Periods per week: **3 (L:0, T:0, P:3)** | Periods per semester: **45** |

**Course Objectives:**

- To impart knowledge on the warp and weft preparation for weaving.
- Calculate the number of machines, production and efficiency for the preparatory section to produce a particular fabric.
- To provide thorough knowledge on analyzing the sample cloths.
- To familiarize handloom weaving and weave different designs in handloom.

**Course Prerequisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| Knowledge of basic Mathematics | | Mathematics 1 | 1 |
| Basic Knowledge of textile mechanics lab | | Basic textile mechanics lab | 2 |

**Course Outcomes:**

On completion of the course, the students will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Identify the warp and weft preparatory machines for weaving | 8 | Understanding |
| CO2 | Demonstrate the preparation of handlooms suitable for various designs and weave the same. | 12 | Applying |
| CO3 | Demonstrate the sample cloths analyzing. | 13 | Applying |
| CO4 | Estimate the production, efficiency of the preparatory section. | 9 | Applying |
| | Lab Exam | 3 | |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 1 | | | | | | |
| **CO2** | 3 | | | | | | |
| **CO3** | 3 | | | | | | |
| **CO4** | 3 | | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

On completion of the course, the student will be able to:

| Module Outcomes | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| **CO1** | **Identify the warp and weft preparatory machines for weaving.** | | |
| M1.01 | Familiarize with the pirn winding machine and Prepare pirns in high speed pirn winder | 2 | Applying |
| M1.02 | Familiarize with the cone winding machine and Prepare cones in cone winder | 2 | Applying |
| M1.03 | Describe the working of sectional warping and mill warping machine mill warping machine. | 2 | Understanding |
| M1.04 | Familiarize with the sizing & beaming machine and Prepare size mixture for given warp. Prepare weaver's beam for the given warp | 2 | Understanding |
| **CO2** | **Demonstrate the preparation of handlooms suitable for various designs and weave the same.** | | |
| M2.01 | Identify the important parts of Handloom, Sketch the lever shedding mechanism, Set the lever shedding mechanism for different weaves | 2 | Understanding |
| M2.02 | Demonstrate the drawing and denting of warp | 1 | Applying |
| M2.03 | Demonstrate the construction of plain design and its derivatives | 3 | Applying |
| M2.04 | Demonstrate the construction of twill designs and its derivatives | 3 | Applying |
| M2.05 | Demonstrate the construction of the toweling fabrics | 3 | Applying |
| | Lab Exam - I | 1.5 | |

| | | | |
|---|---|---|---|
| **CO3** | **Demonstrate the sample cloths analyzing** | | |
| M3.01 | Analyze the fabrics of plain & its derivatives for the following details like Ends & Pricks / Unit space, Count of Warp & Weft, crimp of warp & weft, material of warp & weft and design. | 3 | Applying |
| M3.02 | Analyze the fabrics of Twill its modification and toweling fabrics for the following details like Ends & Pricks / Unit space, Count of Warp & Weft, crimp of warp & weft, material of warp & weft and design. | 4 | Applying |
| M3.03 | Identify the draft, draw the design, lifting plan & denting plan and give the loom requirements to weave the plain & its derivatives. | 3 | Understanding |
| M3.04 | Identify the draft, draw the design, lifting plan & denting plan and give the loom requirements to weave the Twill design & its derivatives and toweling fabrics | 3 | Understanding |
| **CO4** | **Estimate the production, efficiency of the preparatory section** | | |
| M4.01 | Determine the counts of heald and reed | 2 | Understanding |
| M4.02 | Calculate the production and efficiency of the machines in the weaving preparatory section | 4 | Applying |
| M4.03 | Calculate the number of machines required to produce the given quantity of a particular fabric | 3 | Applying |
| | Lab Exam –II | 1.5 | |

The students should keep separate records for

1. Fabric Manufacture,
2. Cloth samples (minimum 30 samples)
3. Design & structure of fabrics and submit certified & bonafide records at the time of practical examination.

**Text / Reference:**

| T/R | Author/Book Title |
|---|---|
| R1 | Thomas W.Fox - The Mechanism of Weaving - Universal Publishing corp., Mumbai |
| R2 | A.Ormerod - Modern Preparation and Weaving Machinery - Woodhead publishing Ltd., England |
| R3 | R.Sengupta - Weaving Calculations - D.B.Taraporevala sons & co Ltd |

**Online Resources:**

| Sl.No | Website Link |
|---|---|
| 1 | http://www.nptel.ac.in/courses |
| 2 | http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html |
| 3 | https://www.textileschool.com |
| 4 | http://textilelearner.blogspot.in |
| 5 | https://textilestudycenter.com |

**SCHEME OF EVALAUATION OF END SEMESTER EXAMINATION**

| Sl.No | Criteria | Marks | Weightage |
|---|---|---|---|
| 1 | Design and procedure of the problem | 15 | 30% |
| 2 | Implementation and output of the problem | 20 | 40% |
| 3 | Viva Voce | 10 | 20% |
| 4 | Record | 5 | 10% |
| | TOTAL | **50** | |